

Anatomy of the Neck

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KEYWORDS

• Neck anatomy • Plastic surgery • Neck lift

KEY POINTS

- Surgery of the neck is commonly performed by Plastic Surgeons.
- Neck anatomy and the location of important structures should be well known prior any neck surgery.
- Risks of complications are increased with secondary and tertiary surgeries.

Editor Commentary: Garrett Wirth was the first resident to rotate with me in the new plastic surgery training program at the Aesthetic and Plastic Surgery Institute at the University of California, Irvine. I asked him to write about neck anatomy because I knew that he would take a dry subject and bring it to life. His important contribution sets the stage for safely treating the aging neck by understanding what structures lie beneath the skin. The danger zones can be avoided after reading and digesting this important subject.

THE NECK AND ITS DIVISIONS

The neck is the region of the body between the clavicle and the mandible. It contains several vital structures (described later) and serves to separate the head from the torso. This discussion begins by describing the divisions of the neck and their contents. Subsequently, the layers of the neck are discussed and, finally, relevance with regards to surgery. Naturally, risks of complications are increased with secondary and tertiary surgeries.

Generally, the neck is clinically divided into an anterior triangle and posterior triangle. The anterior triangle of the neck is bordered by the following:

Anteriorly: the midline of the neck from the sternal notch to the chin

Posteriorly: the anterior margin of the sternocleidomastoid muscle

Superiorly: the mandible's inferior border

Anterior

The anterior triangle is further divided into 4 smaller triangles as follows:

Submental triangle (also known as suprahyoid triangle) is bordered by the hyoid bone

inferiorly, anterior belly of the digastric posterior/anteriorly, and the midline of the neck anteromedially. Its floor is formed by the mylohyoid muscle. This triangle contains submental lymph nodes and veins that subsequently join to form the anterior jugular veins. These veins are only at risk with very deep dissection; thus, injury is unlikely.

Submandibular triangle (also known as the digastric or submaxillary triangle) is bordered by the anterior and posterior bellies of the digastric muscle inferoanteriorly and inferoposteriorly, respectively. The superior aspect is bordered by the lower border of the mandible. This triangle contains the submandibular gland, facial vein and artery branches, and marginal mandibular branch of the facial nerve. The location of these structures deep to the platysma is significant during any neck operation. The location of the gland is relevant to any rejuvenation procedure involving the face/neck. The facial vein and artery must be protected for obvious reasons and damage to the marginal mandibular nerve has obvious deficits.

Carotid triangle (also known as the superior carotid triangle) is bordered by the anterior

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